# Final Project Report

# Multi-Agent Hospitality System

## Course Name: Agentic AI

***Institution Name:*** Medicaps University – Datagami Skill Based Course

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## Problem Statement & Objectives

### 1.1 Problem Statement

Travel planning is an essential part of any journey, but it is often a complex and time-consuming process. Travelers are required to search for suitable destinations, identify popular attractions, organize daily activities, estimate expenses, and structure their plans in a meaningful way. This process typically involves browsing multiple websites, comparing information from different sources, and manually compiling a travel itinerary. As a result, the overall experience becomes inefficient, fragmented, and sometimes overwhelming, especially for users who lack prior travel planning experience.

One of the major challenges in traditional travel planning is the lack of personalization and structure. Users often receive generic recommendations that are not tailored to their specific needs such as trip duration, budget constraints, or personal preferences. Additionally, there is no unified system that combines research, planning, budgeting, and formatting into a single workflow. This leads to incomplete or poorly organized itineraries that may not be practical or optimized for real-world travel scenarios.

### 1.2 Project Objectives

The primary objectives of the **Multi-Agent AI Hospitality Travel Planner** are as follows:

- **To develop an intelligent travel planning system:**
  The project aims to design and implement an AI-powered system that can automatically generate structured travel itineraries based on user inputs such as destination, number of days, and budget preference, thereby reducing manual effort and improving efficiency.
- **To implement a multi-agent architecture:**
  The system utilizes multiple specialized AI agents, each responsible for a specific task such as research and reviewing. This approach enhances modularity, accuracy, and scalability of the overall system.
- **To generate structured and user-friendly outputs:**
  One of the key objectives is to produce clear, organized, and easy-to-understand day-wise itineraries that can be directly used by travelers without requiring further modification.
- **To build an interactive and accessible web interface:**
  The system is designed with a user-friendly interface using Streamlit, allowing users to easily input their travel details and receive instant results, making the application accessible to a wide range of users.

### 1.3 Scope of the Project

The scope of the **Multi-Agent AI Hospitality Travel Planner** defines the boundaries, capabilities, and extent of the system in terms of functionality, usability, and future adaptability. The project focuses on automating travel planning through a structured, intelligent, and scalable approach. The key scope areas are as follows:

- **Travel Itinerary Generation:**
  The system is designed to automatically generate complete and structured travel itineraries based on user inputs such as destination, number of days, and budget preference. It eliminates the need for manual planning by organizing activities into a day-wise format, ensuring clarity, usability, and practical relevance for **Automated** travellers.

- **Multi-Agent AI-Based Processing:**
  The project implements a multi-agent architecture where different AI agents perform specialized tasks such as research, planning, budgeting, content writing, and reviewing. This division of responsibilities improves accuracy, modularity, and scalability, making the system more efficient compared to traditional single-model approaches.

- **Interactive Web-Based User Interface:**
  The application provides a user-friendly interface built using Streamlit, allowing users to input travel details and instantly receive results. The system ensures ease of interaction, accessibility, and real-time response generation, making it suitable for both technical and non-technical users.

- **Integration with External APIs and Tools:**
  The system integrates with external services such as the Groq LLM API for intelligent text generation and DuckDuckGo search for gathering real-world travel data. This ensures that the generated itineraries are context-aware, informative, and aligned with actual travel conditions and attractions.

- **Scalability and Future Expansion:**
  The architecture of the system is designed to support future enhancements without requiring major structural changes. The project can be extended to include features such as hotel booking, flight integration, personalized recommendations, multilingual support, and deployment on cloud platforms, making it adaptable for real-world applications.

## Proposed Solution

### 2.1 Key Features

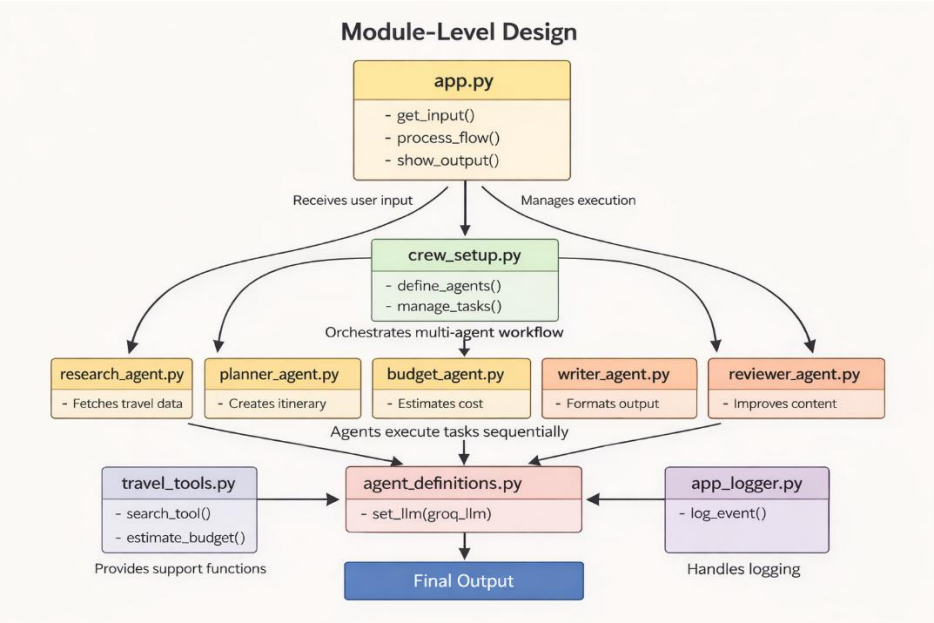
- The system generates structured travel itineraries based on user inputs such as destination, number of days, and budget. This reduces manual effort and provides a clear travel plan.
- The application uses a multi-agent AI architecture where each agent performs a specific task such as research, planning, budgeting, and formatting, improving efficiency and accuracy.
- The system integrates a Large Language Model (LLM) through the Groq API to generate context-aware and meaningful travel content.
- The Research Agent uses external search tools to gather information about tourist attractions and local activities, ensuring practical and relevant outputs.
- The system includes a Budget Agent that estimates travel costs based on duration and budget category, helping users plan expenses effectively.
- The application provides a Streamlit-based interface for easy user interaction and quick generation of travel plans.
- The Writer and Reviewer Agents ensure that the final output is well-structured, readable, and easy to understand.
- The system allows users to download the generated itinerary for offline use and future reference.
- The application follows a modular design, making it easy to maintain and extend with new features.
- The system includes basic logging and error handling to ensure smooth execution and reliability.

### 2.2 Overall Architecture / Workflow

The **Multi-Agent AI Hospitality Travel Planner** follows a modular and layered architecture where each component is responsible for a specific task. The system is designed to ensure clear separation of concerns, scalability, and efficient execution of the travel planning process. Each functional unit is implemented in a separate module, allowing easy maintenance and future enhancements.

### System Components with File Names

- **Main Controller – app.py**
  Handles user input through the Streamlit interface and manages the overall execution flow of the system.
- **Agent Orchestration – crew_setup.py**
  Defines all agents and tasks, and controls the sequential execution of the multi-agent pipeline.
- **Agent Definitions – agent_definitions.py**
  Configures all AI agents and connects them with the Groq LLM for generating responses.
- **Utility Module – travel_tools.py**
  Provides supporting functionalities such as search operations and budget estimation.
- **Logging Module – app_logger.py**
  Maintains system logs for tracking events and handling errors.



## System Workflow

The system operates through a structured sequence of steps as follows:

**Step 1: User Input (Handled by app.py)**

- The user enters travel details such as destination, number of days, and budget through the Streamlit interface.
- The input is received by the main controller (app.py).
- The controller initiates the workflow by passing inputs to the CrewAI orchestrator.

**Step 2: Research Phase (Handled by Research Agent)**

- The Research Agent gathers information about tourist attractions and local experiences.
- It may use external search tools to retrieve relevant and up-to-date data.
- The collected information is passed to the next agent for further processing.

**Step 3: Itinerary Planning (Handled by Planner Agent)**

- The Planner Agent uses the research data to create a structured day-wise travel itinerary.
- It ensures that the plan is organized according to the number of days provided by the user.
- The generated itinerary is forwarded to the Budget Agent.

**Step 4: Budget Estimation (Handled by Budget Agent)**

- The Budget Agent calculates the estimated travel cost based on the selected budget category and trip duration.
- It provides a rough financial overview to help users plan their expenses.
- The output is passed to the Writer Agent.

**Step 5: Content Formatting (Handled by Writer Agent)**

- The Writer Agent combines research data, itinerary, and budget details into a structured and readable format.
- It ensures that the output is clear, concise, and user-friendly.
- The formatted content is passed to the Reviewer Agent.

**Step 6: Output Refinement (Handled by Reviewer Agent)**

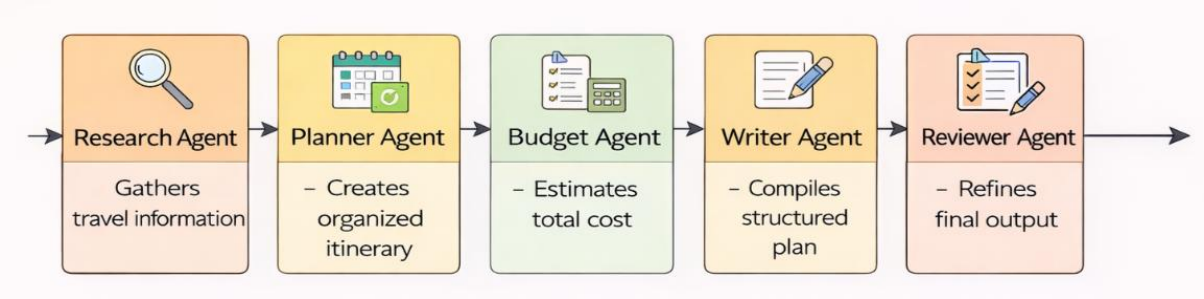
- The Reviewer Agent improves the clarity, readability, and presentation of the final itinerary.

- It ensures consistency and removes any ambiguity in the output.

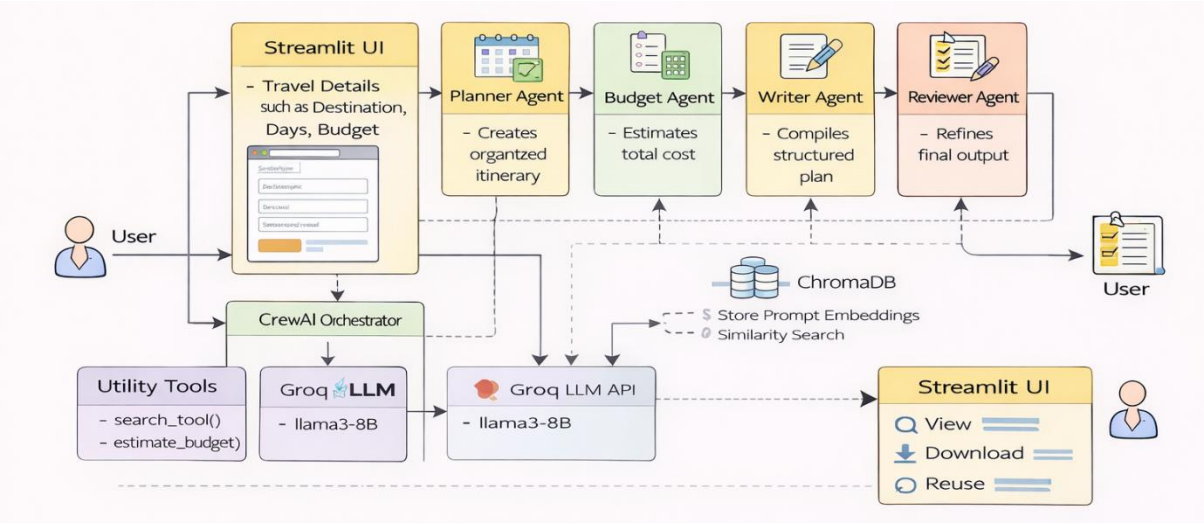
**Step 7: Output Display (Handled by app.py)**

- The final itinerary is returned to the main controller.
- The result is displayed to the user through the Streamlit interface.
- Users can also download the generated itinerary for offline use.

**Agent Pipeline Diagram:**




**End-to-End Workflow Representation:**



### 2.3 Tools & Technologies Used

**Programming Language:** Python

**AI Framework:** CrewAI

**LLM API/Model for Text Generation:** Groq API / llama-3.1-8b-instant

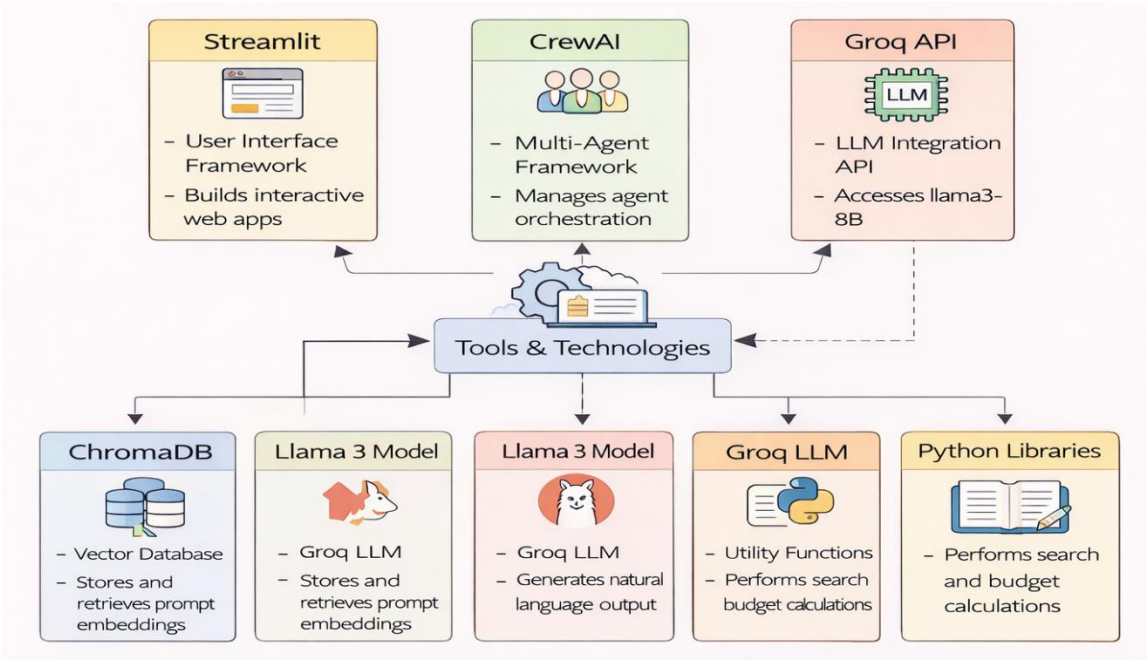
**Search Tool/API:** DuckDuckGo Search

**Web UI Framework:** Streamlit

**Environment Management:** python-dotenv

**Logging Mechanism:** Python Logging Library

**Development Environment:** Visual Studio Code



## Results & Output

### 3.1 Results & Outputs

The **Multi-Agent AI Hospitality Travel Planner** was successfully implemented and tested with different user inputs. The system generates structured and meaningful travel itineraries based on the provided destination, number of days, and budget preference.

- **Input Handling:**
  The system accepts user inputs such as destination, number of days, and budget through the Streamlit interface. The inputs are validated and passed to the multi-agent system for processing.
- **Generated Travel Itinerary:**
  The system produces a day-wise structured itinerary that includes tourist attractions and activities. Each day is clearly organized, making the plan easy to follow and practical for real-world travel.
- **Attractions and Recommendations:**
  The Research Agent successfully retrieves relevant tourist places and local highlights. The generated content is context-aware and aligned with the selected destination.
- **Budget Estimation:**
  The system provides an estimated travel cost based on the selected budget category and trip duration. This helps users plan their expenses effectively.
- **Structured Output Formatting:**
  The final output is formatted in a clean and readable manner using the Writer and Reviewer Agents. The itinerary follows a consistent day-wise structure.
- **Downloadable Output:**
  The application allows users to download the generated travel plan as a text file. This enables offline access and future reference.
- **System Performance:**
  The system responds within a reasonable time and successfully coordinates multiple agents to generate the final output. Minor delays may occur due to API response time.

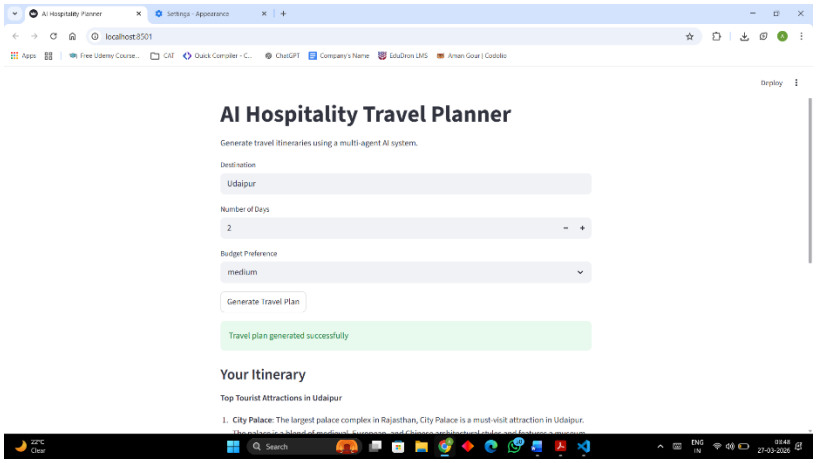
## Sample Execution

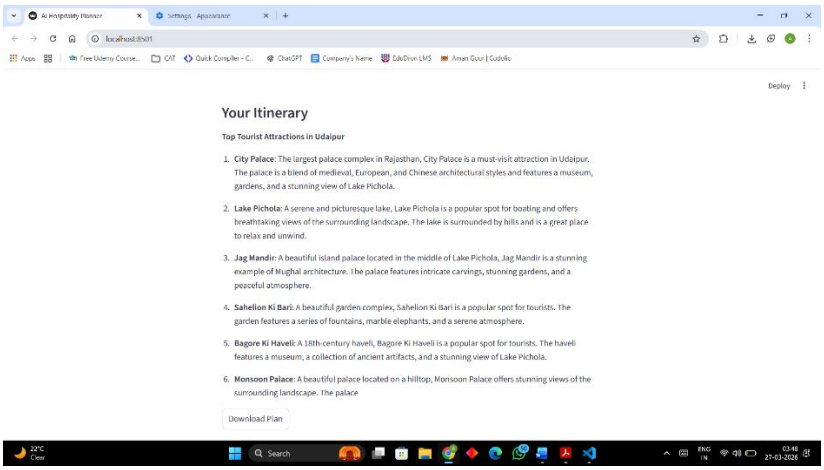
**Input Prompt** : “Destination: Udaipur, Days: 3, Budget: Medium”

**System Output :**

- A structured day-wise travel itinerary for 3 days was generated.
- Relevant tourist attractions and local highlights were included in the plan.
- An estimated travel budget based on the selected category was provided.
- The final itinerary was formatted clearly and displayed on the user interface.
- The generated travel plan was available for download as a text file.

## Screenshot of Outputs:



## 3.2 Reports / Dashboards / Models

### System Architecture Validation

The system follows a **modular multi-agent architecture**, where each component is responsible for a specific task. The main application (app.py) acts as the controller that interacts with the CrewAI orchestrator and manages user input/output.

As described in the High-Level Design, the system is divided into multiple layers:

- The **Streamlit UI** collects user inputs and displays the itinerary
- The **CrewAI Orchestrator** manages task execution and agent coordination
- The **Agent Layer** consists of five specialized agents handling different responsibilities
- The **AI Layer (Groq LLM)** powers all agents with intelligent responses
- The **Utility Layer** provides logging and external data fetching

This modular design ensures **high cohesion and low coupling**, making the system easy to debug, maintain, and extend.

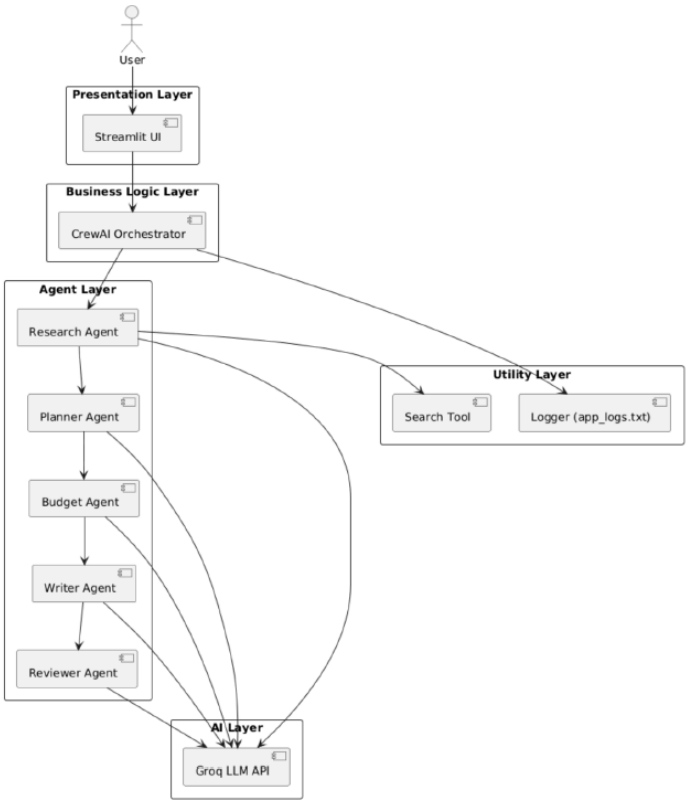



*Figure 1: High Level Architecture Diagram*

## Runtime Sequence Diagram

The runtime execution follows a **sequential agent pipeline**, where each agent processes the output of the previous one.

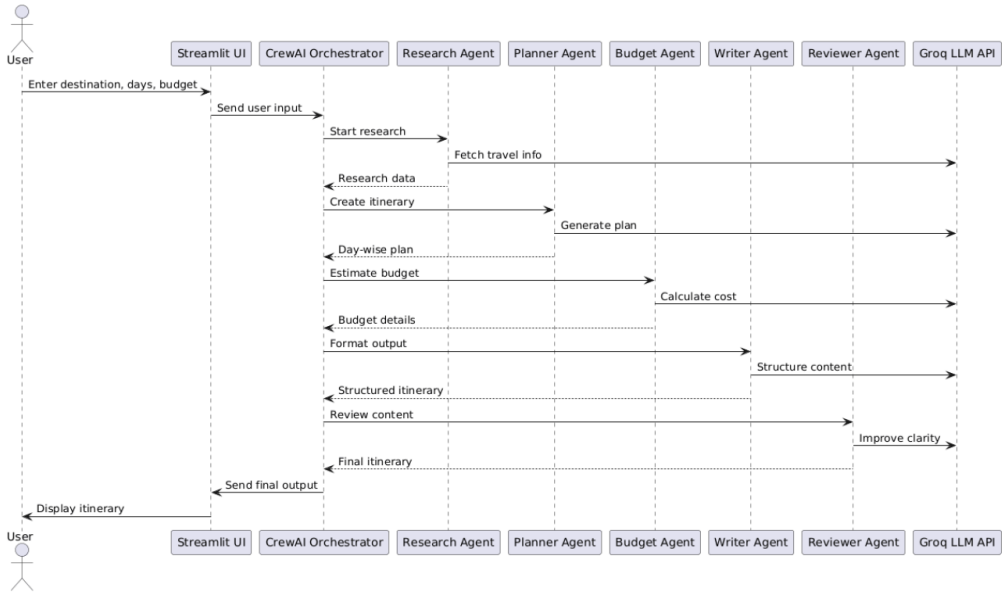
According to the process flow (LLD, page 5) :

User → Streamlit UI → CrewAI Orchestrator → Research Agent → Planner Agent → Budget Agent → Writer Agent → Reviewer Agent → Final Output

### Execution Flow Explanation

- The user provides destination, number of days, and budget
- The orchestrator triggers the pipeline
- Each agent performs a specific task:
  - Research Agent → gathers travel data
  - Planner Agent → creates itinerary
  - Budget Agent → estimates cost
  - Writer Agent → formats output
  - Reviewer Agent → improves clarity
- The final refined itinerary is returned to the UI

This step-by-step execution ensures **progressive refinement of output** and better accuracy.



## Model Details

The system integrates AI services for intelligent text generation and decision-making.

### Text Generation Model

- **Provider:** Groq API
- **Model:** llama-3.1-8b-instant
- **Purpose:**
  - Generate travel recommendations
  - Create structured itineraries
  - Estimate budget
  - Format and refine output

All agents use the same model to maintain **consistency and coherence** across outputs.

## External Data Integration

### Search Tool

- **Provider:** DuckDuckGo Search API
- **Purpose:**
  - Fetch real-time travel information
  - Provide attractions and food suggestions
  - Enhance research agent output

This ensures that the system produces **context-aware and realistic itineraries**.

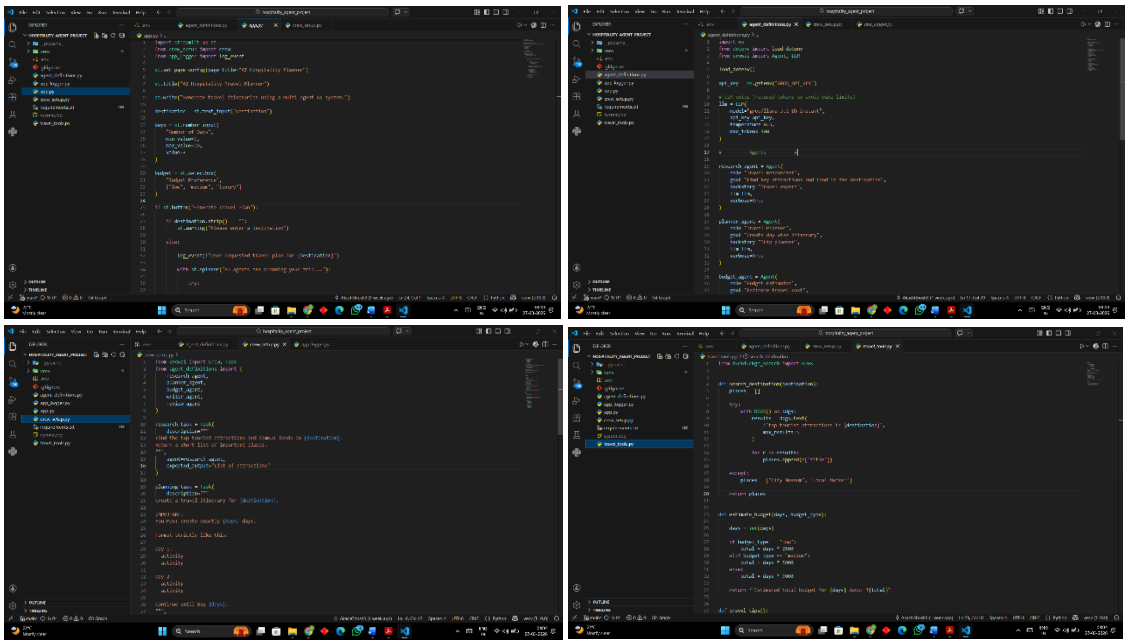
## Code Implementation Evidence

The project was implemented using a **modular Python structure**, where each component is separated into independent files:

- app.py → Main application and UI handling
- crew_setup.py → Defines agent workflow and task pipeline

- agent_definitions.py → Configures all AI agents and LLM
- travel_tools.py → Handles search and budget estimation
- app_logger.py → Manages logging functionality

Each module was developed and tested independently before integration, ensuring **clean architecture and maintainability**.



### 3.3 Key Outcomes

- Successfully designed and implemented a **multi-agent AI system** where specialized agents collaborate to generate a complete travel itinerary, demonstrating the effectiveness of agent-based architecture over single-model approaches.
- Developed a **structured and automated itinerary generation pipeline**, ensuring that outputs follow a consistent day-wise format, improving readability and usability for end users.
- Achieved **modular and scalable system design**, allowing individual components (agents) to be modified, debugged, or extended independently without affecting the entire system.
- Integrated **real-time data retrieval using external search tools**, enhancing the quality and relevance of travel recommendations provided by the system.
- Implemented **budget estimation logic based on user preferences**, enabling personalized travel planning aligned with different financial constraints (low, medium, luxury).

- Demonstrated effective use of **LLM-powered agents for task specialization**, where each agent focuses on a specific responsibility such as research, planning, budgeting, writing, and reviewing.
- Built an interactive **Streamlit-based user interface**, making the system easy to use and accessible without requiring technical expertise.
- Ensured **output refinement through sequential agent processing**, resulting in more accurate, structured, and polished travel plans.
- Incorporated **logging and error handling mechanisms**, improving system reliability, traceability, and ease of debugging during execution.
- Showcased practical understanding of **Agentic AI concepts, orchestration frameworks (CrewAI), and API integration**, making the project aligned with real-world AI system design practices.

## Conclusion

The Multi-Agent Hospitality Travel Planner successfully demonstrates the practical implementation of an agent-based AI system for solving real-world problems like travel planning. By dividing the overall task into smaller responsibilities handled by specialized agents, the system achieves better structure, clarity, and efficiency compared to traditional single-model approaches. The use of a layered architecture ensures clear separation between the user interface, business logic, AI processing, and utility components, making the system modular and maintainable. Integration with the Groq LLM API enables intelligent and context-aware responses, while the inclusion of tools like search and logging enhances reliability and usability. Overall, this project highlights the effectiveness of Agentic AI concepts, orchestration frameworks like CrewAI, and modular design principles in building scalable and user-friendly AI applications.

In addition, the project provided valuable hands-on experience in designing and integrating AI-driven systems using modern tools and frameworks. It strengthened understanding of multi-agent collaboration, prompt structuring, and API-based development, while also emphasizing the importance of modular design and system scalability. This project serves as a strong foundation for building more advanced, real-world AI applications in the future.

## Future Scope & Enchancements

- Integrate **hotel, flight, and transport booking APIs** to provide end-to-end travel planning.
- Add a **database and user authentication system** to store user history and preferences.
- Implement **real-time pricing and availability tracking** for more accurate budget estimation.
- Introduce **parallel agent execution** to reduce response time and improve performance.
- Enhance the UI with **maps, route visualization, and interactive itinerary views.**
- Add **multi-language support** for generating itineraries in different languages.
- Incorporate **recommendation systems** for personalized suggestions (restaurants, events, activities).
- Enable **voice-based input and conversational interface** for better user experience.
- Deploy the application on **cloud platforms** for wider accessibility and scalability.
- Extend the system with **additional agents** (weather agent, safety advisor, local guide agent).